PROJECT DATA SHEET

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JOB/PROJECT : NOSS ON DART MARINA, DARTMOUTH

JOB/PROJECT NO : C8829 **SHEET NO :** COVER

GENERAL :

CLIENT : PREMIER MARINAS

ARCHITECT : N/A

AUTHORITY : TORBAY COUNCIL

JOB/PROJECT DESCRIPTION : BRIDGED ACCESS SCAFFOLDING

PROPOSED USE: FOUNDATION SCAFFOLD SPREADER

OVERALL SIZE : AS CALCS **No. OF STOREYS :** N/A

TYPE OF SOIL INVESTIGATION : TRIAL PITS AND BOREHOLES

BEARING STRATA : 150MM COMPACTED 803

TYPE OF FOUNDATION : TIMBER RAILWAY SLEEPERS

MAXIMUM SAFE BEARING PRESSURE 250 kN/m^2

STRUCTURAL MATERIALS USED : OAK GRADE D40

STANDARDS AND CODES USED IN DESIGN :

General actions: Densities, self weight & imposed loads	BS EN 1991-1-1	☑	Steelwork	BS EN 1993 (EC3)	☐
General actions: Actions on structures exposed to fire	BS EN 1991-1-2	☐	Timber	BS EN 1995 (EC5)	☑
Snow Loads	BS EN 1991-1-3	☐	Masonry	BS EN 1996 (EC6)	☐
Wind Loads	BS EN 1991-1-4	☐	Others		☐
Concrete	BS EN 1992 (EC2)	☐			☐
					☐

AUTHORISATION OF CALCULATION SHEETS AND BENDING SCHEDULES :

CALCULATION SHEET NOS.	CHECKED BY	DATE
Sheet nos: F01 - F02 inclusive	9P Aw	11/9/25

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Calculations

Contract	NOSS ON DART MARINA DARTMOUTH	Job No.	C8829	Sheet No.	F01
		Designed by	A.C.M.	Date:	OCT 25

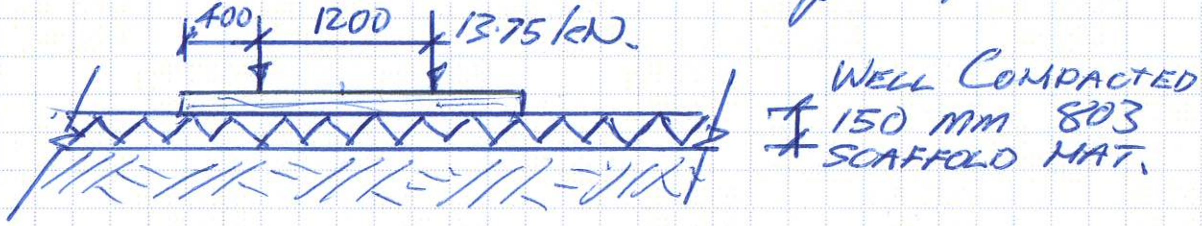
BRIDGED ACCESS SCAFFOLD FOUNDATION

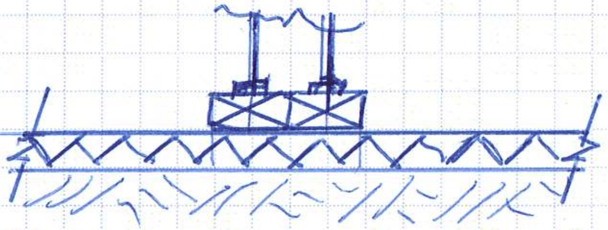
Consider foundation spreader.

From Optima Drg. No. 25/OPT/22101-001 A load at end of bridge beams = 55kN.

This is shared over 4 No. Standards ie. 13.75 kN/standard.

Consider 2 No. standards on 2.0m long sleeper.





Contact stress @ U/s of sleepers = $55/(2.0 \times 0.25 \times 2)$

$= 55 kN/m^2 < 250 kN/m^2$ ∴ ok.

with 2:1 spread through 803.

Stress @ underside of stone = $55/(2.15 \times 0.65)$

$= 39 kN/m^2 < 60 kN/m^2$ ∴ ok.

Consider timber.

UDL on 1 sleeper = $\left(\frac{55}{2}\right)/2.0$ = 13.75 kN/m char

= 19.25 kN/m ult.

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Calculations

Contract	NOSS ON DART MARINA DARTMOUTH	Job No.	C8829	Sheet No.	F02
		Designed by	A.C.H.	Date:	OCT 25

$M_d = 19.25 \times 1.2^2/8 = 3.46\,kNm$

or $= 19.25 \times 0.4^2/2 = 1.54\,kNm.$

σ_b reqd $= 3.46 \times 10^6 \Big/ \left(\frac{250 \times 125^2}{6}\right) = 5.31\,N/mm^2.$

for D40 oak sleeper in Service class 3.
permissable bending stress $= \underset{k_{mod}}{0.65} \times 30 = 19.5\,N/mm^2.$

$\gg 5.31\,N/mm^2$

$\therefore$ ok.

$\therefore$ USE 2.0m LONG × 250wd × 125 dp D40 OAK SLEEPER BELOW BRIDGE BEAM SUPPORT STANDARDS. (2No.).